Before moving on, stop the application with Ctrl-C and all the containers with `docker-compose down` (from polar-deployment/docker).

## 11.6 Testing Spring Security and OpenID Connect

The importance of writing autotests is usually apparent to developers. Still, things can get challenging when it comes to security, and it sometimes ends up not being covered by automated tests because of its complexity. Fortunately, Spring Security provides several utilities to help you include security in your slice and integration tests in a simple way.

In this section, you'll learn how to use the `WebTestClient` support for Spring Security to test OIDC authentication and CSRF protection. Let's get started.

### 11.6.1 Testing OIDC authentication

In chapter 8, we tested REST controllers exposed by Spring WebFlux by relying on the `@SpringWebFlux` annotation and `WebTestClient`. In this chapter, we added a new controller (`UserController`), so let's write some autotests for it with different security setups.

First of all, open your Edge Service project, create a `UserControllerTests` class annotated with `@WebFluxTest(UserController.class)` in src/test/java, and autowire a `WebTestClient` bean. So far, the setup is similar to what we used in chapter 8: a slice test for the web layer. But we need some extra setup to cover security scenarios, as shown in the following listing.